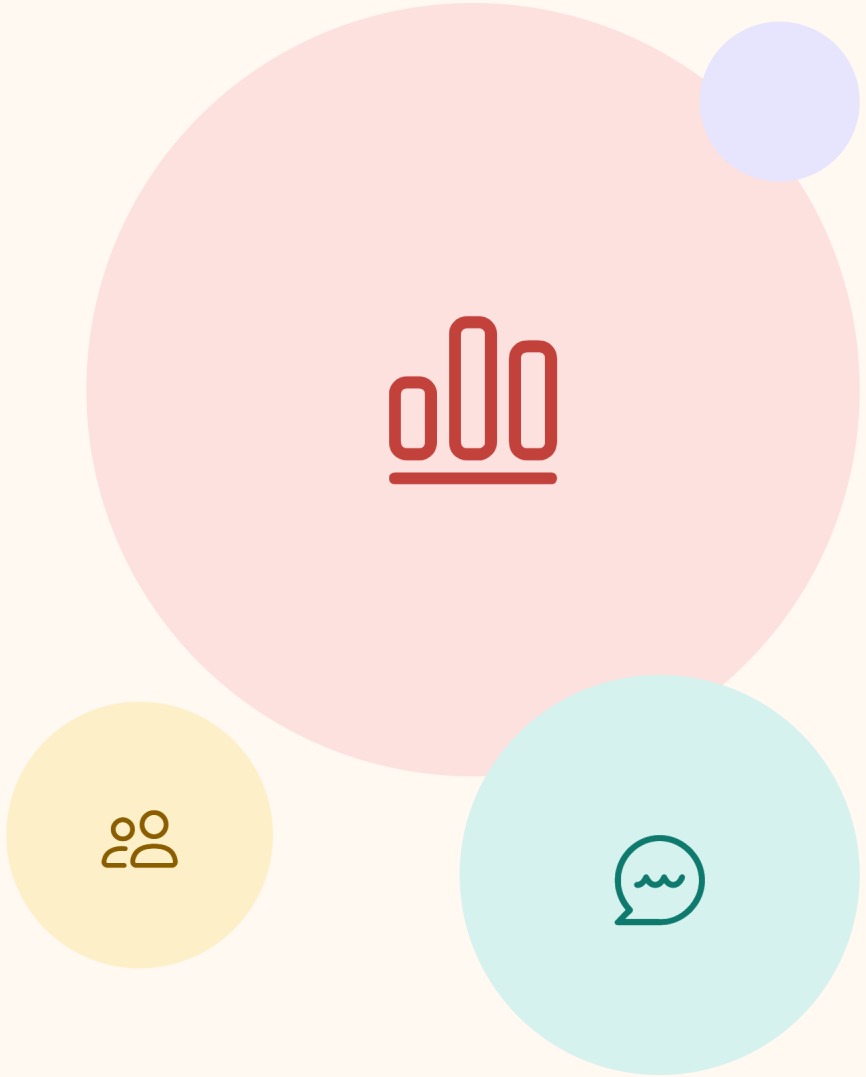
CUSTOMER SUPPORT ANALYSIS

What support tickets reveal about unhappy customers and churn

Two years of helpdesk data from Flowbase, five business questions, and what the support team can do next.

Julia Silva · Data analysis portfolio project · October 2026



THE CONTEXT

Two years of Flowbase's support desk in one dataset

17,381
support tickets

3,200
customer accounts

35
support agents

24
months, Jul 2024 to Jun 2026

Flowbase is a made-up B2B software company. The data is **synthetic**: I generated it with a Python script, with realistic patterns and noise, and hid two business events in it, a platform outage and a price increase.

WHY IT MATTERS

Many customers leave, and most leave in silence



26%

of customers churned in two years (833 of 3,200)



62%

of them left without giving any reason (513 of 833)

So support tickets are one of the few early warning signs the business has.

THE BUSINESS QUESTIONS

Five questions I set out to answer

1 · SPEED	How fast is the team, and how often is the SLA missed?	
2 · SATISFACTION	What makes customers give low CSAT scores?	
3 · EMOTION	What is the emotion in customer messages?	Next step
4 · CHURN	Who leaves, and can we predict it?	
5 · EVENTS	Are the outage and the price increase visible in the data?	

HOW I WORKED

Clean, explore, predict

1. Clean

Every missing value had a reason, and all 19 logic checks passed. Survey skipping looked random.

2. Explore

Eight analysis questions with tables and charts, comparing categories, priorities and periods.

3. Predict

A logistic regression that flags customers at risk of churn, using only data from before they left.

Tools: Python · pandas · NumPy · matplotlib · seaborn · scikit-learn · Jupyter

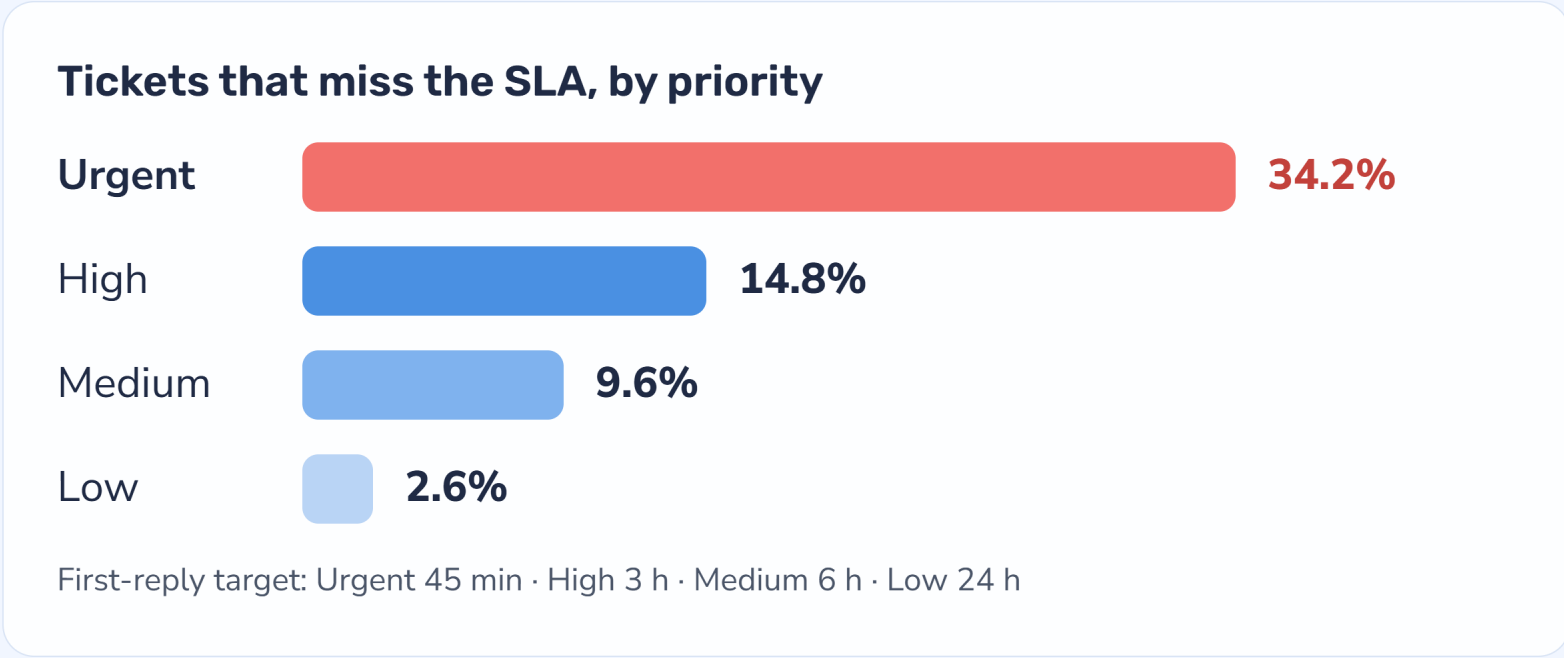
1 · SPEED

The team is fast on average, but Urgent tickets are most often late

59.5 min
median wait for a first reply

14 hours
median time to resolve a ticket

11.2%
of all tickets miss the SLA



First-reply target: Urgent 45 min · High 3 h · Medium 6 h · Low 24 h

1 · SPEED

Two places where time is lost

BUG REPORTS

3.2 days

The slowest category to resolve (median). A third of bug tickets are escalated, because the fix depends on Engineering, not on the support team.

CANCELLATION REQUESTS

1.8 hours

The longest wait for a first reply of any category, while these customers are deciding whether to leave. A missed chance to keep them.

2 · SATISFACTION

A late reply makes customers five times as likely to be unhappy

First reply was late

On time

5.5%

Late

28.9%

About 5× as many unhappy

Ticket was reopened

Not reopened

6.3%

Reopened

28.3%

About 4.5× as many unhappy

Escalated to Tier 2

Not escalated

7.6%

Escalated

12.0%

Only 1.6×, a small effect

Share of customers who gave a CSAT score of 1 or 2 · based on 6,874 survey answers

2 · SATISFACTION · THE TWIST

Performance looked like a product problem, but it was a speed problem

After cancellations, Performance had the most unhappy customers (13.8%). I expected the slow product to be the cause, but splitting the scores by SLA showed something else:

Unhappy Performance customers (CSAT 1 or 2)



29.6% of these tickets missed the SLA, almost 3× the average (11.0%).

5 · EVENTS · OUTAGE, 11 MARCH 2025

One outage, one day, 268 tickets



≈ 11×

a normal day, almost all of it Performance tickets

52.0%

of Performance tickets missed the SLA (19.4% on normal days)

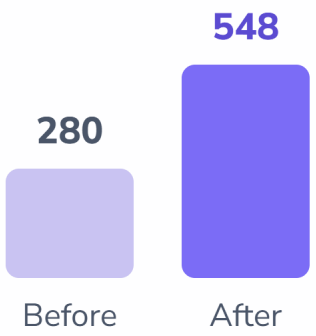
25.2%

unhappy customers, up from 9.9% on normal days

5 · EVENTS · PRICE INCREASE, 1 SEPTEMBER 2025

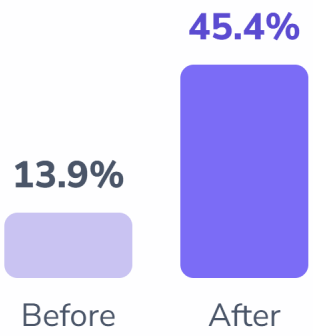
After the price increase, CSAT stayed good, but customers left

Billing tickets



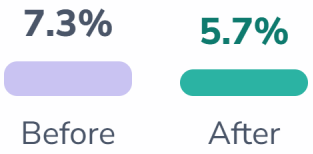
Almost doubled

Price questions



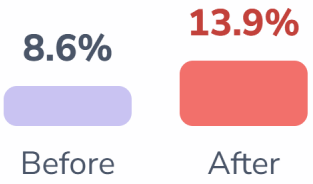
Of all Billing tickets

Unhappy (CSAT 1–2)



Slightly better

Churn in 90 days



About 1.6× higher

⚠ **CSAT said everything was fine. Churn said otherwise.**

3 months before vs after

4 · CHURN · WHO LEAVES

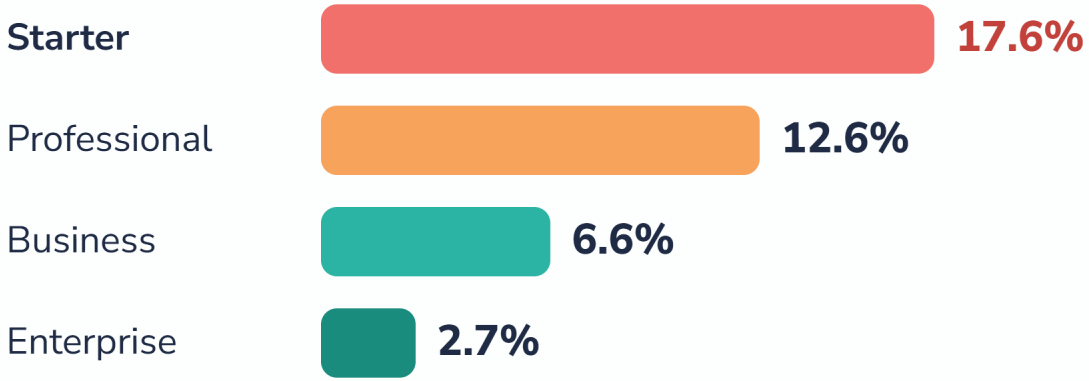
Billing tickets and small plans are the clearest churn warnings

Churn within 90 days after a ticket



The highest rate after Cancellation Requests, and the largest share of all tickets followed by churn (17.4%).

After a Billing ticket, by plan



4 · CHURN · PREDICTION

A simple model finds 6 in 10 customers who are about to leave

167 test customers who really churned



■ 103 caught (61.7%) ■ 64 missed

1.6×

more churners in its list than in a random pick

0.70

ROC-AUC: moderate, clearly better than guessing

Strongest warning signs

- ⚠ Contacts support often
- ⚠ Starter or Professional plan
- ⚠ Pays monthly
- ⚠ Never answers surveys
- ⚠ Many Billing tickets

✓ **Lower risk:** pays more, Enterprise

WHAT TO DO

Five recommendations

1	Answer **Urgent** and **Performance** tickets first, especially during incidents.	Support
2	Solve problems properly the first time, so tickets don’t get reopened.	Support
3	Treat **Billing** tickets as an early churn warning, especially after a price change.	Customer Success
4	Reply faster to **Cancellation Requests**, now the slowest first reply.	Support
5	Use the churn model to build a weekly check-in list.	Customer Success

HONEST NOTES

Limitations and next steps

⚠ Limitations

- The data is synthetic, so patterns are cleaner than in a real company.
- Only 39.5% of tickets have a CSAT score.
- Before/after comparisons show a link, not proof of cause.
- The model is moderate, because most reasons for leaving are not in support data.

Next steps

- Text analysis (NLP) on customer messages to find each message’s emotion: question 3.
- Compare the model with others, such as a random forest.
- Build a dashboard from the clean data, with alerts for ticket spikes.

THE TAKEAWAY

Fast, first-time fixes keep customers happy. Billing tickets show who might leave.

Protect the SLA

Fix it the first time

Follow up on Billing

Thank you · Julia Silva